Hard Derivation 3: Common Capacity comparative static

1. Common coverage

$$F = V_F \rho D + R$$

$$\Delta(A) = \sum_J \mu_J (1 - \psi_J(A)) D_J(A)$$

$$\phi(A) = \min\left[1, \frac{F}{\Delta(A)}\right]$$

For $\Delta(A) > 0$ and $0 < F < \Delta(A)$

$$\phi(A) = \frac{F}{\Delta A}$$

then

$$\frac{\partial \phi(A)}{\partial F} = \frac{1}{\Delta A}$$

If $0 < \Delta(A) < F$, the coverage is saturated and the derivate is zero

So, out of break points

$$\phi_F(A) = \frac{1 \mid 0 < F < \Delta(A)}{\Delta(A)}$$

With the convention $\phi = 0$ when $\Delta = 0$

2. Marginal deficit-cost coefficient

Remember

$$a_K = i\psi_K + K(1-\phi)(1-\psi_K)$$

Derivative with respect to F

$$a_{K,F} = K(1-\psi_K)\frac{\partial(1-\phi)}{\partial F}$$

$$= -K(1-\psi_K)\phi_F$$

Then

$$a_{K,F} = -K(1-\psi_K)\frac{1 \mid 0 < F < \Delta(A)}{\Delta(A)}$$

Increasing common capacity lowers the marginal cost associated with an uncovered deficit

3. Define the first-order condition

$$G(\mu_K, F) = J'_K(\mu_K, F)$$

thus

$$G(\mu_K, F) = r - E_A[a_K(F)\pi_K^- + b_K \pi_K^+]$$

For an inner solution

$$G(\mu_K^*(F), F) = 0$$

4. Partial derivatives

$$G_\mu = J''_\mu$$

$$G_F = -E_A[a_{K,F}\pi_K^-]$$

Replacing $a_{K,F}$

$$G_F = KE_A\left[1_{|0<F<\Delta(A)|}\frac{(1-\psi_K)\pi_K^-}{\Delta(A)}\right]$$

$$\boxed{G_F \geq 0}$$

5. Implicit Function Theorem

As

$$G(u_k^*(F), F) = 0,$$

Since $G(u_0, F_0) = 0$, $G_u(u_0, F_0) > 0$ and G is continuously differentiable locally, the IFT guarantees a differentiable stationary branch $u^*(F)$ near F_0

Derivative with respect to F

$$G_u \frac{\partial u_k^*}{\partial F} + G_F = 0$$

$$\frac{\partial u_k^*}{\partial F} = \frac{-G_F}{G_u}$$

Replacing

$$\frac{\partial u_k^*}{\partial F} = \frac{K E_A\left[1_{\{0<F<D(A)\}} \frac{(1-\psi_k)(\pi_k)}{D(A)}\right]}{J''(u_k^*)}$$

6. Sign and conditions

$$J''_k(u_k^*) > 0$$

and, if exists a set of positive probability states where:

$$F < D(A), \quad \psi_k < 1, \quad \pi_k > 0$$

then

$$G_F > 0$$

Therefore

$$\frac{\partial u_k^*}{\partial F} < 0$$

Regular interior optimum; positive curvature; fixed market schedules, shock law and domain; no relevant coverage kink; differentiability and no global-minimum switch.

Finally

$$F = v_F p D + R$$

With v_F, p and D fixed

$$\frac{\partial u_k^*}{\partial R} = \frac{\partial u_k^*}{\partial F}$$

More common credits reduces optimal private self-insurance under a conditional best response not necessarily in general equilibrium.